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SBU 363 01: Money and Capital Markets

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<u>Introduction:</u>

In general, interest rates affect financial markets across all levels, but one sets the foundation: the risk-free rate. The Determinants of Risk-Free Rate are mainly affected by inflation expectations and monetary policy. This interest rate is observed in financial markets, including the U.S. Treasury bill, and is a combination of the real risk-free rate and inflation premium. In capital markets, it is crucial to understand what affects and drives the risk-free rate and how nominal and real values branch off from each other.

This paper asks what macroeconomic forces drive the risk-free rate and compares the nominal vs. real rates over time? This question is relevant, especially considering the current economic environment. Between 2022 and 2023, the FED rapidly raised interest rates post-pandemic. Congress.gov states, “Since July 2023, the Fed has maintained a target range of 5.25%-5.5%, the highest target since 2001” (Congressional Research Service, 2024). To respond to the highest inflation in four decades, the FED raised federal funds by 525 basis points. As a result, this pushed treasury bills to the highest level in over two decades. CNBC stated, “The 10-year Treasury yield was last up 11 basis points to 4.793, after climbing to 4.8% earlier on Tuesday. The 30-year Treasury yield rose as high 4.924%, also the highest since 2007” (Min & Kiderlin, 2023). Additionally, the Treasury Inflation-Protected Securities (TIPS) from the Federal Reserve Economic Data (FRED) database showed real rates were negative through 2021, but had a sharp climb with nominal rates (Federal Reserve Bank of St. Louis, n.d.). After inflation hit a peak, the FED cut rates by a cumulative 175 basis points through 2024 and 2025. Federal Reserve Board stated, “In support of its goals and in light of the shift in the balance of risks, the Committee decided to lower the target range for the federal funds rate by 1/4 percentage point to 3-1/2 to 3-3/4 percent” (Board of Governors of the Federal Reserve System, 2025). However, at the FED’s April meeting, they held rates due to inflation pressures from rising energy costs. This shows how rapidly macroeconomic factors can shift the risk-free rate outlook.

Section two will go over the loanable funds theory and the Fisher equation. Section three will cover the macroeconomic drivers of the risk-free rate and present data from FRED and the World Bank. Section four discusses the implications of recent rate movements and inflation.

Background and Theory:

Two theories that help explain the risk-free rate are the loanable funds theory and the Fisher equation. The loanable funds theory is the theory of interest rate equilibrium based on the supply and demand for loanable funds. The textbook states, “A theory of interest rate determination that views equilibrium interest rates in financial markets as a result of the supply of and demand for loanable funds” (Saunders et al., 2024). The supply of loanable funds rises, and real rates decline when households save money and government borrowing declines. The supply of loanable funds decreases when the demand for them increases, which makes real interest rates rise. In simpler terms, when the amount of available money shrinks, there is more competition to borrow the money.

The Fisher effect was named after economist Irving Fisher, and it is the relationship between the real risk-free rate and the nominal risk-free rate. The textbook states,

> “The Fisher effect theorizes that nominal risk-free rates observed in financial markets (e.g., the one-year Treasury bill rate) must compensate investors for (1) any reduced purchasing power on funds lent (or principals lent) due to inflationary price changes and

(2) an additional premium above the expected rate of inflation for foregoing present consumption…" (Saunders et al., 2024).

This shows how lenders want compensation and money for the time value of money, but also for their purchasing power that decreased. This relationship can be observed through TIPS, the nominal and real yield, and the breakeven inflation rate, which is a market-derived measure of inflation expectations. Treasury Direct states, "They were first auctioned in January 1997 after the market expressed a strong interest in the inflation-indexed asset class" (TreasuryDirect, n.d.). Before TIPS, the real rate was undetectable in real time. Nowadays, this is a standard tool in monetary policy.

Analysis:

The macroeconomic factors that contribute to the risk-free rate can be seen in both the changes to the monetary policy that is implemented by the Federal Reserve, as well as in the changes to the yields of the U.S. Treasury notes over time. One of the main influences upon the risk-free rate is the ability of the Federal Reserve to control both short-term and long-term interest rates within the economy.

The Federal Reserve, along with other central banks across the globe, have the power to control the interest rates that are applied to short-term loans and instruments. As a result, their decisions regarding the monetary policy that they implement can impact the expected interest rates that are applied to all periods along the yield curve. For instance, according to the U.S. Bank, the Fed implemented interest rate increases of 525 basis points between early 2022 and 2023 (U.S. Bank, 2026). These increased interest rates were implemented as a means of countering the high inflation rates within the U.S. economy, leading to an increase in the risk-free rates for those U.S. Treasury notes.

Data from the Federal Reserve Economic Data (FRED) helps to illustrate the impact of each of these macroeconomic factors. For instance, during the pandemic, the 10-year interest rate for U.S. Treasury notes dropped to rates of between 0.5% and 1.0% (Federal Reserve Economic Data \[FRED\], 2026). However, after the Federal Reserve began to increase the interest rates of U.S. treasuries to combat the high inflation rates within the nation, the 10-year interest rates for those notes increased as well; they rose to rates of between 4.0% and 4.5% between 2023 and 2025 (Federal Reserve Economic Data \[FRED\], 2026). After inflation rates began to decline within the U.S. economy, however, the Federal Reserve began to adjust the interest rates of the U.S. Treasuries. According to the U.S. Bank, following the decline of the core PCE inflation rates within the U.S. to rates of 3% in September of 2024, the Federal Reserve began to implement interest rate cuts for the United States; the rates were lowered to rates of 3.5% - 3.75% by early 2026 after implementing cuts of 175 basis points to the interest rates (U.S. Bank, 2026).

Table 1

10-Year Nominal Treasury Yield vs. TIPS Real Yield (Approximate Annual Averages, 2020–2024)

Year	Nominal (DGS10)	Real TIPS (DFII10)	Breakeven Inflation
2020	0.89%	−0.97%	1.86%
2021	1.45%	−1.04%	2.49%
2022	2.95%	0.53%	2.42%
2023	3.96%	1.84%	2.12%

2024	4.20%	2.10%	2.10%

Note. Source: FRED (DGS10, DFII10); U.S. Department of the Treasury.

The risk-free rate is influenced by various economic factors. The actions of the FED had an impact upon the changes to the interest rates within the United States. Between 2022 and 2023, the FED began to raise the interest rates within the United States by 525 basis points. Following that period, however, the FED began to lower the interest rates within the nation, implementing 175 basis points of interest rate cuts between 2024 and 2026 (U.S. Bank, 2026). Data from the Federal Reserve Economic Data and Table 1 illustrate the changes to the 10-year interest rates and the real TIPS interest rates from 2020 until 2024. The 10-year interest rates began at 0.89% in 2020, but increased to 4.20% in 2024. Additionally, the real TIPS interest rates began at -.97% in 2020 and increased to 2.10% in 2024. Thus, the table helps to illustrate the change in both nominal and real interest rates within the United States over this period. Furthermore, the table indicates that the real interest rates began in a negative range in 2020 but became positive over time. Additionally, the breakeven inflation rate has remained relatively stable between the range of 2.1% to 2.4% since 2022.

Discussion:

When the risk-free rate goes up or down, it has an impact on the entire financial world. The risk-free rate is the rate that is provided by the United States government as the safest bet in the market. Thus, anyone else in the financial world, from common investors to banks and policymakers, must change their behavior and their financial decisions when the rate changes.

The risk-free rate acts as a benchmark for all those who invest or start up a business. When the risk-free rate is low, such as in 2020 when safe investments like Treasury bonds paid very little interest, investors are forced to take risks with their money in order to achieve better returns. Thus, they place their money into the stock and real estate markets. When the risk-free rate is high, as in 2024 when investors could receive a safe four to five percent return on their money by lending it to the United States government, they require higher returns from stocks before they are willing to place their money at risk with company stocks. Thus, the stock market, especially for the riskier tech companies, will experience a drop in their stock prices. When the risk-free rate increases, the cost of borrowing money increases for the companies that require loans in order to start new businesses or products. Because of the high interest rates, many companies will delay their projects, which will impact the country's economy overall.

Banks are impacted by these rate changes, especially if those changes are rapid. For example, a problem that they face is the bond problem. Because of the rapid increase of the risk-free rate between 2022 and 2023, the banks lost value in their holdings of older, low-paying government bonds. Finally, banks also have to engage in what is referred to as the deposit fight. The goal of the banks was to discourage their customers from moving their money out of their bank's savings accounts into the high-yield government funds. To combat this, banks have to pay higher interest rates on the deposits of their customers, which costs the bank money.

Finally, the Federal Reserve must balance the rate of the risk-free interest rates. When inflation was high between 2022 and 2023, the Federal Reserve increased the risk-free interest rates to slow the economy down and reduce the inflation. The ultimate goal of the Federal Reserve during this time was to find a soft landing in the economy. This would mean that the inflation was slowly decreasing without causing a massive recession and unemployment in the United States. Between 2024 and 2026, however, the Federal Reserve made it very difficult to find such a soft landing. The interest rates that the Fed set for the risk-free rate were lowered in an attempt to encourage economic activity. However, energy and gas prices spiked between 2025 and 2026, which caused the Federal Reserve to pause its rate cuts. Thus, there does not seem to be a perfect interest rate for the risk-free market, and what is considered to be the perfect rate today may change overnight to another rate.

Conclusion:

In conclusion of our discussion of the risk-free rate, it becomes evident that this statistic is not a static figure that exists within the economy at any given time. Instead, the risk-free rate is a statistic that is primarily driven by the inflation rate of the country, the monetary policy that is established for that nation, and the supply and demand for the loanable funds within the nation. The Fisher equation helps to establish the relationship between the nominal interest rates of loans, the real interest rates, and the inflation rate that is common within a given country. Additionally, the loanable funds theory establishes how the interest rates within a given nation are influenced both by the supply of those loanable funds, as provided by the citizens of the nation through their savings, and the demand for those loanable funds from entities like the government that wish to borrow from those citizens. Between 2020 and 2025, each of these factors began to impact the risk-free rate of the United States. For instance, during the COVID-19 pandemic, the economic activity in the United States created suppressed yields for 10-year treasury notes, which fell to values of as low as 0.89% (Federal Reserve Bank of St. Louis, n.d.). As a result of inflation rates rising to levels not seen in 40 years, however, the Federal Reserve implemented 525 basis points of increases to the federal funds rate between 2022 and 2023, leading to risk-free rates of 10-year notes of 4.20% (Congressional Research Service, 2024) and real rates of 2.10% by 2024 (Federal Reserve Bank of St. Louis, n.d.).

However, following the peak of the inflation rates within the economy, 175 basis points of interest rate reductions were made between early 2025 and early 2026, as well as the additional reduction of the federal funds rate at the Federal Reserve's April 2026 meeting for the nation's economy (U.S. Bank, 2026).

The movement of the risk-free rate of the United States has an impact upon the country's investors, financial institutions, and policymakers. For the investors of the nation, changes in the risk-free rate influence the required return rates for each of the asset classes within the investor's portfolio. For the nation's financial institutions, changes in the risk-free rate influence their lending rates to individuals and businesses, the values of their existing bonds, and their interest in borrowing from their savers. Finally, policymakers that regulate the Federal Reserve and the nation's economy must find a way to adjust the risk-free rate and the federal funds rate to minimize inflation within the economy and the United States economy's potential downturn. Despite the movements of the 10-year treasury yields and the federal funds rate between 2020 and 2025, however, the breakeven inflation rate remained stable between 2.10% and 2.49% throughout this time period (Federal Reserve Bank of St. Louis, n.d.). Thus, despite the risk-free rate's relationship with inflation, the determination of its influences remains one of the most essential questions for those within the financial world to understand about the various economies across the world.

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